

Advanced Java Programming (MCS-313) Final Assignment

Develop a **Spring Boot** application on **any application domain of your choice**. The project should demonstrate the concepts and technologies covered in the course.

Minimum Requirements

The application must include the following features:

- Complete **CRUD (Create, Read, Update, Delete)** operations.
- Layered architecture using **Controller, Service, Repository, and Entity**.
- Integration with a relational database using **Spring Data JPA**.
- Input validation and proper exception handling.
- Authentication and authorization (where applicable).
- Search, filtering, sorting, and pagination.
- User-friendly interface or well-documented REST APIs.
- Proper project structure following Spring Boot best practices.
- Clean, well-documented, and maintainable code.

Additional Features

In addition to the mandatory CRUD functionality, the application must implement **at least three** additional features. Examples include:

- File upload/download
- Email notification
- Dashboard and reports
- Data visualization
- PDF/Excel/CSV export
- Audit logging
- Activity history
- Role-based access control
- REST API integration
- Caching
- Scheduled tasks
- Any other relevant feature that demonstrates advanced Spring Boot concepts.

Note: Students are free to choose **any application topic or domain**. The evaluation will be based on the implementation of the required features, code quality, application design, and adherence to Spring Boot best practices, **not on the choice of application domain**.

Marking Scheme

Advanced Java Programming (MCS)

Assignment Marking Scheme (100 Marks)

S.N.	Evaluation Criteria	Marks
1	Problem Identification and Application Design	10
2	CRUD Functionality (Create, Read, Update, Delete)	20
3	Database Design and Spring Data JPA Integration	10
4	Application Architecture (Controller, Service, Repository, Entity)	10
5	Input Validation and Exception Handling	10
6	Authentication and Authorization	10
7	Additional Features (Minimum Three)	15
8	User Interface / REST API Design and Usability	5
9	Code Quality, Documentation, and Best Practices	5
10	Project Demonstration and Viva	5
Total		100

Evaluation Guidelines

1. Problem Identification and Application Design (10 Marks)

- Appropriate application scope and objectives
- Well-designed database schema and application flow

2. CRUD Functionality (20 Marks)

- Complete implementation of Create, Read, Update, and Delete operations
- Correct business logic and data handling

3. Database Design and Spring Data JPA Integration (10 Marks)

- Proper entity relationships
- Efficient database design

- Correct use of Spring Data JPA

4. Application Architecture (10 Marks)

- Clear separation of Controller, Service, Repository, and Entity layers
- Modular and maintainable code structure

5. Input Validation and Exception Handling (10 Marks)

- Server-side validation
- Meaningful error messages
- Proper exception handling mechanisms

6. Authentication and Authorization (10 Marks)

- Secure login mechanism
- Role-based access control (if applicable)
- Protection of application resources

7. Additional Features (15 Marks)

Implementation of at least **two** advanced features beyond CRUD, such as:

- Search, filtering, sorting, or pagination
- File upload/download
- Email notifications
- Reports or dashboards
- PDF/Excel/CSV export
- Audit logging
- REST API integration
- Caching
- Scheduled tasks
- Any other advanced Spring Boot feature

8. User Interface / REST API Design and Usability (5 Marks)

- Clean and intuitive interface or well-designed REST APIs
- Consistent navigation and response structure

9. Code Quality, Documentation, and Best Practices (5 Marks)

- Clean, readable, and maintainable code
- Proper naming conventions
- Comments where necessary
- README with setup and execution instructions

10. Project Demonstration and Viva (5 Marks)

- Successful demonstration of all implemented features
- Ability to explain application design, implementation, and technologies used

Resources Required for Submission

Students must submit the following items:

1. **Complete Spring Boot Project Source Code**
 - Well-organized project with all source files.
 - Include all necessary configuration files (e.g., `application.properties` or `application.yml`).
2. **Database**
 - SQL script (`.sql`) containing database schema and sample data, **or**
 - Embedded database (e.g., H2) configured with the project.
3. **Project Report (PDF)**

The report should include:

 - Project Title
 - Problem Statement
 - Objectives
 - System Architecture
 - Database Design (ER Diagram/Class Diagram)
 - Application Features
 - Screenshots of the Application
 - Technologies Used
 - Conclusion
4. **README File**
 - Project description
 - Software requirements
 - Steps to configure and run the application
 - Default login credentials (if applicable)
5. **Presentation Slides (PPT/PDF)**
 - Overview of the project
 - Features implemented
 - Architecture and database design
 - Demonstration screenshots
 - Future enhancements
6. **Project Demonstration**
 - Live demonstration of the working application during the evaluation/viva.
 - Students should be prepared to explain the implementation, application architecture, and technologies used.

Submission Format

- Submit all files in a single compressed folder (.zip or .rar).
- Name the submission using the following format:
RollNo_StudentName_ProjectTitle.zip

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